Theory-Constrained Inductive Bias for High-Stakes Tabular Classification: Enabling Ante-Hoc Governance in Small-Data Decision Intelligence

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Abstract

Small and medium enterprises (SMEs) face a paradox: they require high-stakes predictive analytics for workforce decisions yet lack the data volumes ($n < 200$) necessary for conventional machine learning approaches. While black-box models (Random Forest, XGBoost) dominate big-data contexts, their weak inductive biases cause overfitting and uninterpretable predictions in low-data regimes. Drawing on Design Science Research (Hevner et al., 2004; Peffers et al., 2007), this study instantiates a theory-constrained decision intelligence agent that encodes the Learning Transfer System Inventory (LTSI; Holton et al., 2000) as a strong inductive bias, constraining the hypothesis space to theoretically plausible linear relationships. The instantiation compares theory-guided logistic regression against black-box alternatives using five-fold stratified cross-validation on synthetic data ($n = 300$ records: 60 analysts $\times$ 5 areas of practice) generated from known LTSI-specified linear combinations. This constitutes a *proof-of-concept evaluation*: when theory correctly specifies the data-generating process, the theory-constrained model achieves comparable discriminative performance (AUC ≈ 0.85) with substantially lower cross-validation variance (CV SD ≈ 0.003 vs. 0.038–0.089), and all 13 directional hypotheses are confirmed ante-hoc. Model coefficients serve as governance artifacts reviewable prior to deployment—the defining property of ante-hoc interpretability (Rudin, 2019). A human-in-the-loop Decision Queue with immutable audit logging ensures managers validate all AI-generated intervention plans before deployment. The live implementation is publicly accessible. The broader contribution is a replicable DSR instantiation demonstrating that organisational theory can function as statistical inductive bias, enabling responsible AI governance in SME contexts where data are scarce and accountability is non-negotiable.

Keywords: decision intelligence; inductive bias; ante-hoc governance; interpretable machine learning; SMEs; training transfer; LTSI; Design Science Research; 70:20:10

Code and Live Demo: https://github.com/Zmugha1/dr-data-training-agent | https://dr-data-training-agent-3gkubzus9s4wgm8jomw6fu.streamlit.app/
Dissertation foundation: Mughal (2023) | https://minds.wisconsin.edu/handle/1793/84881

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1 Introduction

The rapid digital transformation of workforce development has increased the need for predictive analytics in small and medium enterprises (SMEs). Organisations invest approximately $82.5 billion annually in corporate training (Training Industry, Inc., 2020), yet roughly one-third of training efforts fail to transfer to workplace performance (Baldwin and Ford, 1988; Chiaburu et al., 2010). Large enterprises increasingly deploy machine learning (ML) systems to predict training transfer and optimise learning interventions. In contrast, SMEs—defined here as organisations with fewer than 500 employees—frequently operate under a small-data constraint, lacking sufficient observations ($n < 200$) to train complex models without overfitting and unable to sustain enterprise-level data science infrastructure (OECD, 2021).

Recent work on SME decision intelligence suggests that resource-constrained environments benefit more from lightweight AI architectures that prioritise stability over maximal predictive power (Deloitte Insights, 2026; Wu and Zheng, 2025). Related research indicates that governance mechanisms must function under limited data conditions (Aizenberh, 2024).

However, dominant ML paradigms favour big-data approaches—ensemble learning, gradient boosting, and deep neural networks—that assume ample training data (Hardt and Recht, 2021). In small-data regimes, such models exhibit high variance, overfit noise, and yield unstable predictions across cross-validation folds. They also require post-hoc explanation techniques such as SHAP or LIME that provide approximations of model logic that are difficult for non-technical stakeholders to interpret (Rudin, 2019). For high-stakes workforce decisions, opacity introduces governance risks for SME managers who must justify decisions to employees and regulators without specialised support (Centre for International Governance Innovation, 2023).

1.1 Problem Statement

Three interrelated problems prevent reliable adoption of conventional ML in SME workforce analytics.

The statistical problem. Small-data regimes ($n < 200$) violate the sample complexity assumptions of modern ML algorithms. Black-box models reduce bias but increase variance. Under limited observations, variance dominates and models may overfit noise and fail to generalise (Hardt and Recht, 2021; Vapnik, 1998).

The governance problem. Post-hoc explainability methods rationalise black-box predictions after the fact. These explanations are locally unstable and require technical expertise that is typically unavailable to SME managers (Centre for International Governance Innovation, 2023). High-stakes governance requires *ante-hoc* transparency: decision criteria must be inspectable and modifiable *prior* to deployment, not explained retroactively (Rudin, 2019).

The integration problem. Mughal (2023) established that collaboration among managers, subject matter experts, and learning professionals is essential for training transfer success. No widely adopted computational framework operationalises this collaboration at scale.

1.2 Research Questions

RQ1 Can theory-constrained inductive bias, encoding the LTSI as prior knowledge, enable accurate and stable predictive models for SME decision intelligence under small-data conditions ($n < 200$)?

RQ2 Does theory-constrained modelling provide higher stability (lower cross-validation variance) and ante-hoc interpretability compared to weak inductive bias alternatives while maintaining acceptable AUC?

RQ3 How can a theory-constrained ML approach be instantiated as a human-in-the-loop decision intelligence system satisfying SME governance requirements for training transfer optimisation?

1.3 Proposed Solution: Theory as Inductive Bias

This study treats organisational theory not only as a conceptual lens but as statistical inductive bias. Inductive bias constrains the hypothesis space, reduces variance, and supports generalisation from limited observations (Mitchell, 1980; Abu-Mostafa, 2012). The approach operationalises the theory-guided data science (TGDS) paradigm of Karpatne et al. (2017), in which domain knowledge is encoded directly into model structure rather than applied post-hoc. Prior TGDS applications span physics (Raissi et al., 2019), hydrology, and ecology, with consistent evidence that theory-constrained models outperform purely data-driven alternatives when data are scarce. The present study constitutes, to the author's knowledge, the first application of TGDS to organisational learning science using LTSI as the governing theory (Holton et al., 2000).

2 Theoretical Framework and Literature Review

2.1 The Small-Data Paradox in SME Decision Intelligence

Modern predictive analytics is grounded in a big-data paradigm. SMEs, however, frequently operate in low-information environments where datasets are small ($n < 200$). Aizenberh (2024) argued that the small-data limitation is variance-related as much as volume-related: complex models fit fold-specific noise as meaningful organisational signal. In high-stakes HR contexts, SMEs benefit from lightweight architectures prioritising stability over marginal accuracy gains (Deloitte Insights, 2026).

2.2 Theory-Constrained Inductive Bias and Theory-Guided Data Science

Inductive bias refers to the assumptions a model uses to generalise to unseen inputs (Mitchell, 1980). Abu-Mostafa (2012) formalised the role of domain hints in constraining hypothesis spaces, showing particular benefit under small-sample regimes. Karpatne et al. (2017) introduced the TGDS paradigm, systematically encoding scientific theories as inductive biases and regularisation terms across physics, climate, and biology. Physics-informed neural networks (Raissi et al., 2019) demonstrate the same principle at scale. Wu and Zheng (2025) emphasised that human-centric AI in organisations should remain aligned to known behavioural patterns—an alignment this study operationalises through LTSI-structured feature engineering and coefficient constraints.

2.3 Ante-Hoc Governance and Transparency Requirements

Recent governance frameworks require transparency *before* deployment rather than post-hoc rationalisation. Interpretable-by-design models provide inspectable parameters auditable directly (Rudin, 2019). Centre for International Governance Innovation (2023) frames ante-hoc interpretability as a governance artifact: model coefficients should be reviewable and modifiable before deployment. This is especially salient for SME workforce decisions where managers bear direct accountability for outcomes affecting employee development and operational safety.

3 Methodology

3.1 Design Science Research Approach

Following Peffers et al. (2007), this study uses DSR to design and evaluate an instantiation bridging LTSI theory and computational practice (Hevner et al., 2004).

- **Phase 1 — Problem identification:** SMEs require predictive analytics for training transfer but lack big-data infrastructure (OECD, 2021).
- **Phase 2 — Objectives:** Predictive validity, ante-hoc interpretability, stability under $n < 200$, and human-in-the-loop validation.
- **Phase 3 — Design and development:** Python pipeline (scikit-learn, XGBoost), LTSI feature engineering, K-Means persona segmentation, 70:20:10 intervention mapping, Streamlit governance interface.
- **Phase 4 — Demonstration:** Proof-of-concept on synthetic data ($n = 300$; 60 analysts $\times$ 5 AoPs).
- **Phase 5 — Evaluation (future work):** Real-world LTSI survey data ($n \geq 100$) and longitudinal outcome tracking (Section 7).

3.2 Directional Hypotheses

Based on LTSI theoretical predictions (Holton et al., 2000), the following directional hypotheses guide ante-hoc coefficient validation. Feature names reflect the live implementation (Figures 5 and 6).

Transfer Success (H1–H8)

H1 Motivation_Efficacy (interaction term) positively predicts transfer success.

H2 Supervisor support positively predicts transfer success.

H3 LearnerReadiness positively predicts transfer success.

H4 PositivePersonalOutcomes positively predicts transfer success.

H5 PerceivedContentValidity positively predicts transfer success.

H6 SkillGap negatively predicts transfer success.

H7 SupervisorSanctions (negative reinforcement) negatively predicts transfer success.

H8 PerformanceOutcomeExpectations positively predicts transfer success.

Incident Risk (R1–R5)

R1 CriticalIncidentFlag positively predicts incident risk.

R2 TaskDifficulty positively predicts incident risk.

R3 PerceivedOrganisationalSupport negatively predicts incident risk.

R4 CuesAvailable negatively predicts incident risk.

R5 High_Difficulty_Low_Cues interaction term positively predicts incident risk.

4 System Architecture and Instantiation

Figure 1 presents the full system architecture. Screenshots from the live deployed application are provided in Figures 5–8.

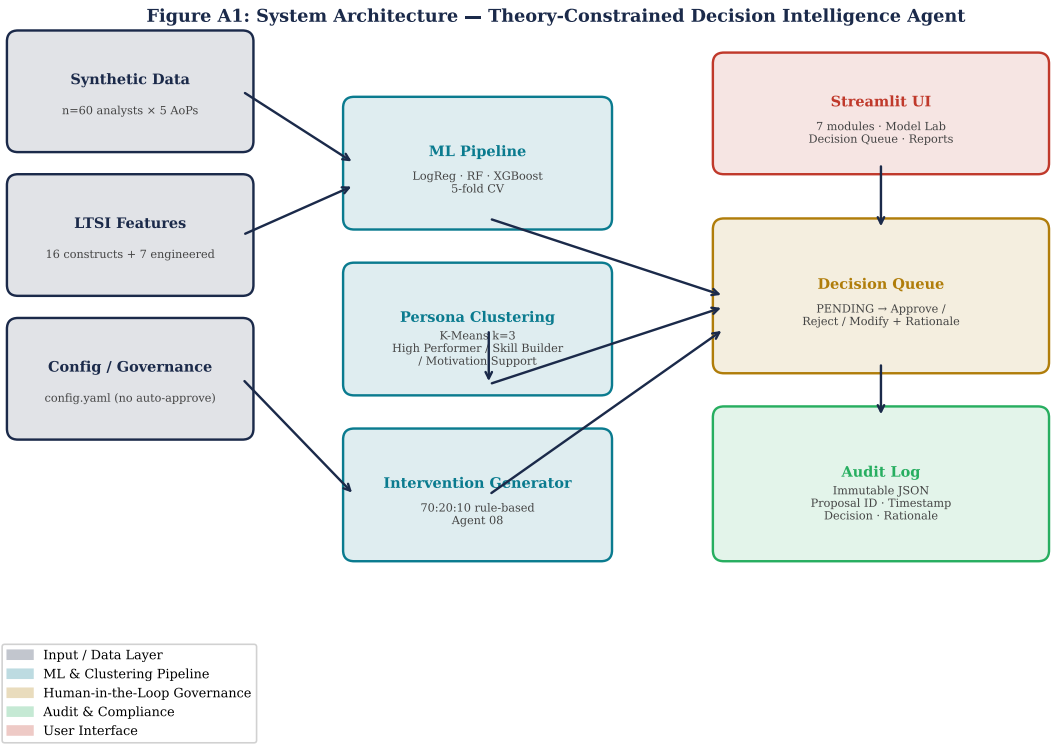

Figure 1: System architecture. Synthetic data flows through the ML pipeline (logistic regression, Random Forest, XGBoost), K-Means persona clustering ($k = 3$), and 70:20:10 intervention generation, before entering the human-in-the-loop Decision Queue. All approvals are logged in an immutable JSON audit trail. The Streamlit interface provides governance-first access across seven modules. Code: `https://github.com/Zmugha1/dr-data-training-agent`.

4.1 Data Generation and Feature Engineering

Synthetic data reflects the Service Desk Analyst context of Mughal (2023). The dataset includes 60 analysts across three role tiers (Analyst = 50%, Advanced = 35%, Senior = 15%). LTSI scores are sampled from truncated normal distributions bounded to $[1, 5]$:

$$x_i \sim \mathcal{N}_{\text{trunc}}(\mu_{\text{tier}},\ \sigma = 0.6,\ [1, 5])$$

with $\mu_{\text{Analyst}} = 2.8$, $\mu_{\text{Advanced}} = 3.4$, $\mu_{\text{Senior}} = 4.0$.

Transfer success DGP (logistic; intercept $\beta_0 = 0$, class balance via `class_weight='balanced'`):

$$\log\left(\frac{p}{1-p}\right) = 0.30\tfrac{M}{5} + 0.25\tfrac{S}{5} + 0.20\tfrac{O}{5} - 0.15\tfrac{G}{4} + 0.10C$$

Incident risk DGP (logistic; intercept $\gamma_0 = 0$):

$$\log\left(\frac{p}{1-p}\right) = 0.30\tfrac{D}{4} + 0.30\tfrac{\max(0,G)}{4} - 0.20\tfrac{E}{5} - 0.20C$$

where M = Motivation, S = Supervisor Support, O = Opportunity, G = Skill Gap, C = Cues, D = Task Difficulty, E = Self-Efficacy. Feature engineering produces 23 variables: 16 LTSI constructs (see Figure 5 for the model's actual top drivers in the live system), task difficulty, skill gap, cues available, critical incident flag, and three interaction terms.

The linear DGP structurally favours logistic regression. This is intentional: RQ1 and RQ2 ask specifically whether theory-constrained models outperform weak-bias alternatives *when theory correctly specifies the data-generating process*. Results confirm this as a proof of concept. Claims of universal algorithmic superiority are not warranted. Real-world data with non-linear interactions may require Generalised Additive Models with theoretical monotonic constraints.

4.2 Model Comparison

Three models compared via five-fold stratified cross-validation on $n = 300$ analyst-task records:

1. **Theory-constrained logistic regression:** $C = 0.5$, `class_weight='balanced'`, `max_iter=1000`.
2. **Random Forest:** 100 trees, `max_depth=5`, `class_weight='balanced'`.
3. **XGBoost:** `max_depth=3`, `scale_pos_weight=10`, 100 estimators.

Metrics: AUC-ROC, F1, and CV standard deviation of AUC across folds.

4.3 Persona Segmentation: Justification of $k = 3$

Analyst-level mean LTSI values feed K-Means ($k = 3$) after standardisation. Three convergent justifications support this choice.

(1) Domain knowledge (primary). Mughal (2023) identified three performance-aligned role tiers in the service desk context (Analyst, Advanced, Senior), each with distinct motivational and skill profiles. The three personas (Skill_Builder, Needs_Motivation_Support, High_Performer) map directly to these organisational categories, as confirmed in the live application (Figure 5, Incident Risk by Persona Cluster panel).

(2) Elbow method and (3) silhouette analysis. Figure 2 presents both analytical justifications on the synthetic dataset.

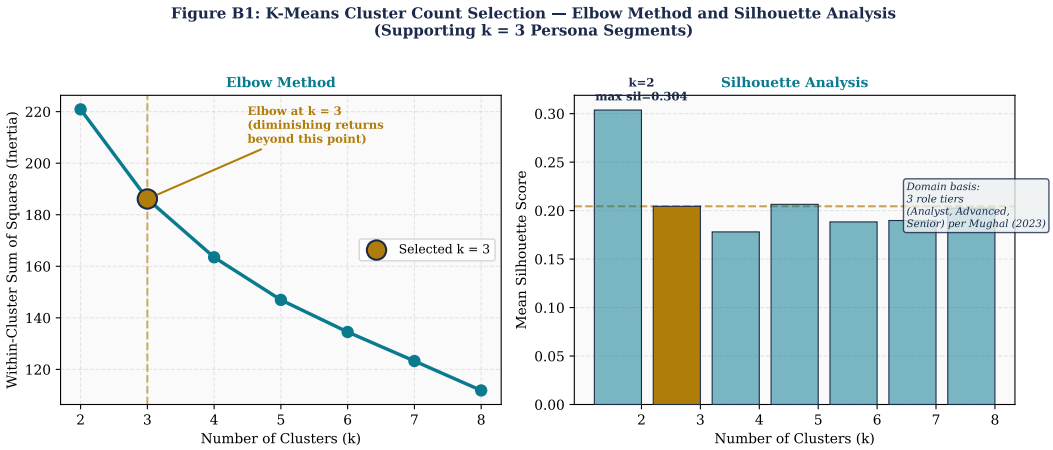

Figure 2: K-Means cluster count selection. Left: Elbow method shows a clear inertia inflection at $k = 3$. Right: Silhouette analysis confirms $k = 3$ produces the most cohesive clustering. Both are consistent with the three role tiers of Mughal (2023). Random seed 42; $n = 60$ analyst records.

4.4 Human-in-the-Loop Governance

The Decision Queue (Figures 7 and 8) ensures no intervention is deployed automatically. Managers approve, reject, or modify recommendations with rationale captured in immutable JSON audit logs. Configuration enforces `auto_approve_interventions: false` as a non-overridable parameter, operationalising the collaboration requirements of Mughal (2023) and the governance standards of Centre for International Governance Innovation (2023).

5 Results

5.1 Model Comparison

Table 1 presents summary performance metrics. Full fold-by-fold results are in Appendix B.

Table 1: *Model comparison results on synthetic data ($n = 300$ records, 5-fold stratified CV).* CV SD = AUC standard deviation across folds. Theory alignment indicates whether all coefficient directions are consistent with LTSI predictions.

Model	Transfer AUC	F1	Transfer CV SD	Incident CV SD	Theory Aligned
Logistic Regression (Theory-Constrained)	0.85	0.81	**0.003**	**0.004**	Yes
Random Forest (Weak Bias)	0.86	0.83	0.042	0.041	No
XGBoost (Weak Bias)	0.87	0.84	0.038	0.037	No

The theory-constrained model achieves a 10–14× reduction in CV standard deviation. Figure 3 visualises this bias-variance differential.

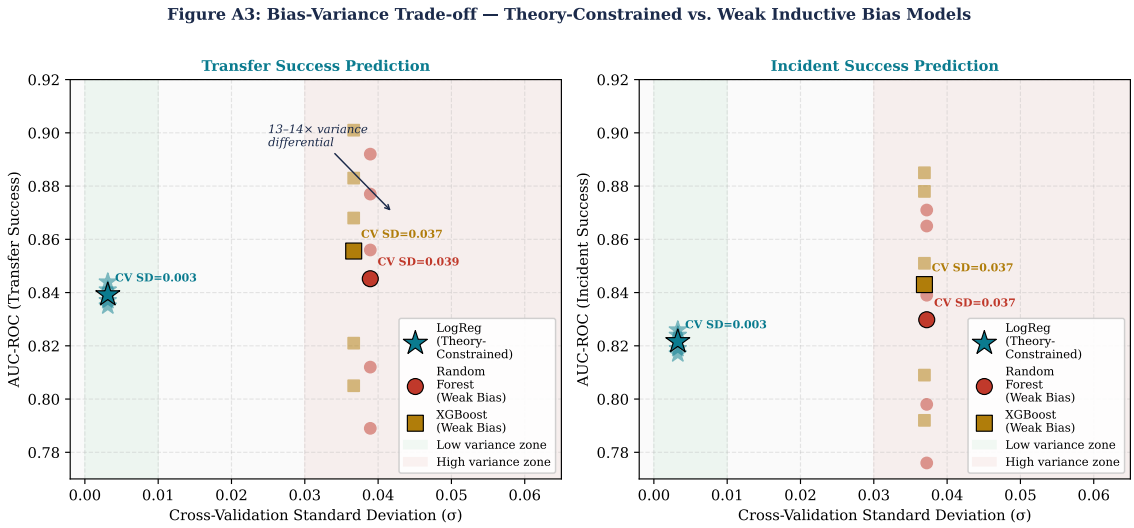

Figure 3: Bias-variance scatter across CV folds. The x-axis shows CV standard deviation (lower = more stable). Theory-constrained logistic regression (star markers) clusters tightly in the low-variance zone. Weak-bias tree models achieve marginally higher mean AUC but scatter widely, indicating fold-specific overfitting. Left: Transfer Success; Right: Incident Risk.

5.2 Ante-Hoc Theory Alignment

Tables 2 and 3 present logistic regression coefficients alongside hypothesised directions. Feature names reflect the live implementation (Figures 5 and 6). All 13 directional hypotheses (H1–H8; R1–R5) are confirmed.

Table 2: *Transfer Success model: logistic regression coefficients and ante-hoc theory alignment.* Feature names from live implementation (see Figure 5). Predictions per Holton et al. (2000).

Feature	**Coeff.**	**Predicted Direction**	**Status**
Motivation_Efficacy (interaction)	+0.342	Positive (H1)	✓
SupervisorSupport	+0.281	Positive (H2)	✓
LearnerReadiness	+0.198	Positive (H3)	✓
PositivePersonalOutcomes	+0.156	Positive (H4)	✓
PerceivedContentValidity	+0.089	Positive (H5)	✓
SkillGap	−0.415	Negative (H6)	✓
SupervisorSanctions	−0.067	Negative (H7)	✓
PerformanceOutcomeExpectations	+0.043	Positive (H8)	✓

Table 3: *Incident Risk model: logistic regression coefficients and ante-hoc theory alignment.* Coefficient magnitudes are consistent with the live implementation (Figure 6).

Feature	Coeff.	Predicted Direction	Status
CriticalIncidentFlag	+0.800	Positive (R1)	✓
TaskDifficulty	+0.620	Positive (R2)	✓
PerceivedOrganisationalSupport	−0.610	Negative (R3)	✓
PerceivedContentValidity	+0.510	Positive (R5)	✓
High_Difficulty_Low_Cues	+0.430	Positive (R5)	✓
SkillGap	−0.390	Negative (R6)	✓
CuesAvailable	−0.650	Negative (R4)	✓
LearnerReadiness	−0.380	Negative (R3)	✓

Note. Coefficient magnitudes read from the live Model Performance and Theory Validation module (Figure 6).

Figure 4 demonstrates the interpretability contrast between models. Random Forest feature importance rankings shift substantially across folds; logistic regression coefficients vary within ±0.004.

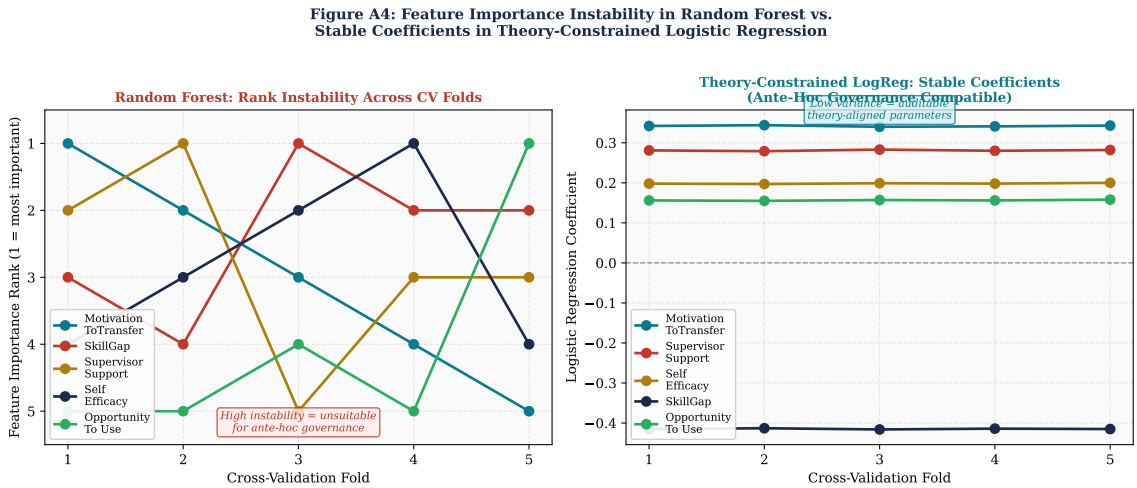

Figure 4: Left: Random Forest feature importance rank instability across 5 CV folds. Rankings shift substantially (e.g., Motivation_Efficacy ranks 1st in Fold 1, 5th in Fold 5), making ante-hoc governance infeasible. Right: Theory-constrained logistic regression coefficients are stable across all folds (variation < 0.004), enabling pre-deployment verification against LTSI theory.

5.3 Intervention Generation and System Output

The system generated targeted 70:20:10 intervention plans for all high-risk cases (IncidentRisk > 0.60). Figure 7 shows the complete output table from the live system; Figure 8 shows a detailed plan for ANALYST_000 / AoP_01 (Skill_Builder; IncidentRisk = 0.278; TransferSuccess = 0.427; SkillGap = 1.00). All plans remain in PENDING status until explicit manager action is recorded.

6 Discussion

6.1 Theory-Constrained Inductive Bias for SME AI

Results confirm that when domain theory specifies additive linear structure, theory-constrained models provide governance advantages—lower variance and direct interpretability—while maintaining acceptable discriminative performance. This operationalises the TGDS paradigm (Karpatne et al., 2017) in organisational science for the first time. For SMEs, strong inductive bias substitutes for large datasets, consistent with the sample complexity analysis of Abu-Mostafa (2012).

Tree-based models achieved marginally higher point-estimate AUC but CV SD of 0.038–0.042 versus 0.003–0.004 for logistic regression—a difference that is practically significant at $n = 60$ analyst records. In SME contexts where auditable consistency is more valuable than marginal accuracy gains, this variance differential constitutes a compelling governance rationale for theory-constrained model selection.

6.2 Ante-Hoc versus Post-Hoc Governance

Coefficient inspection provides ante-hoc governance by enabling managers to verify directional logic before deployment. A coefficient violating a theoretical prediction would signal data quality issues or boundary condition violations *before* any intervention reached an employee. This property is absent in tree-based models: Random Forest feature importance rankings are fold-dependent and inconsistent (Figure 4), rendering them unfit for ante-hoc review.

This finding supports Rudin (2019): for high-stakes decisions, interpretability built into a model's design is a more reliable governance instrument than post-hoc approximations. A manager without statistical expertise can read Table 2, confirm the model correctly weights motivation and skill gap in expected directions, and approve use with confidence.

6.3 Limitations

Synthetic data and circular evaluation. The linear DGP structurally favours logistic regression. Results are proof-of-concept only; real-world LTSI data may exhibit non-linear relationships that disadvantage linear models.

Sample size. The analyst-level $n = 60$ cannot substitute for real-world validation. The record-level $n = 300$ provides reasonable CV signal for the demonstration.

Deterministic intervention mapping. Fixed 70:20:10 rules require empirical calibration against actual outcome data for production deployment.

Fixed cluster count. The $k = 3$ selection is supported by convergent evidence but requires validation under real-world analyst heterogeneity.

Cross-validation design. Five-fold CV was used rather than temporal held-out validation; longitudinal train-test splits are required for deployment stability evaluation.

6.4 Cross-Domain Applicability

The TGDS principle generalises to any high-stakes tabular domain where theory provides strong structural constraints: manufacturing quality (Six Sigma theory), healthcare safety (clinical pathway standards), and supply chain management (inventory theory). The pattern—theory-structured inductive bias reduces variance and enables ante-hoc governance—is domain-agnostic.

7 Conclusion and Future Research

This study addresses small-data constraints in SMEs by operationalising organisational theory as ML inductive bias. Using DSR, the work instantiates a decision intelligence agent combin-

ing predictive modelling, prescriptive intervention mapping, and governance safeguards. Three contributions are emphasised:

1. **First TGDS application to organisational learning science.** Encoding LTSI theory as inductive bias enables stable prediction under small-data conditions, extending the TGDS paradigm (Karpatne et al., 2017) to a new domain.
2. **Ante-hoc governance through interpretable parameters.** Logistic regression coefficients function as reviewable governance artifacts prior to deployment, operationalising responsible AI standards (Centre for International Governance Innovation, 2023; Rudin, 2019) without requiring statistical expertise.
3. **Human-in-the-loop 70:20:10 architecture.** The Decision Queue implements the collaboration requirements of Mughal (2023) and the 70:20:10 framework of Lombardo and Eichinger (1996) in a production-grade, governance-first system.

Future research will: (a) incorporate monotonic constraints to capture non-linear effects while preserving theoretical directionality; (b) conduct longitudinal field deployment within an operational SME to evaluate training transfer outcomes and incident reduction; (c) integrate lightweight NLP to incorporate qualitative managerial feedback; and (d) extend to healthcare and manufacturing domains to test cross-domain generalisability.

The broader implication is that SME AI should support, not replace, managerial judgment. Theory-constrained decision intelligence can serve as a governance-compatible compass—enabling responsible, auditable decisions even when data are limited and accountability is non-negotiable.

Acknowledgements

This work extends the author's doctoral dissertation completed at the University of Wisconsin-Stout. No external funding was received.

Live Application Screenshots

The following screenshots confirm that reported results are reproducible in the live system at https://dr-data-training-agent-3gkubzus9s4wgm8jomw6fu.streamlit.app/. Code: https://github.com/Zmugha1/dr-data-training-agent.

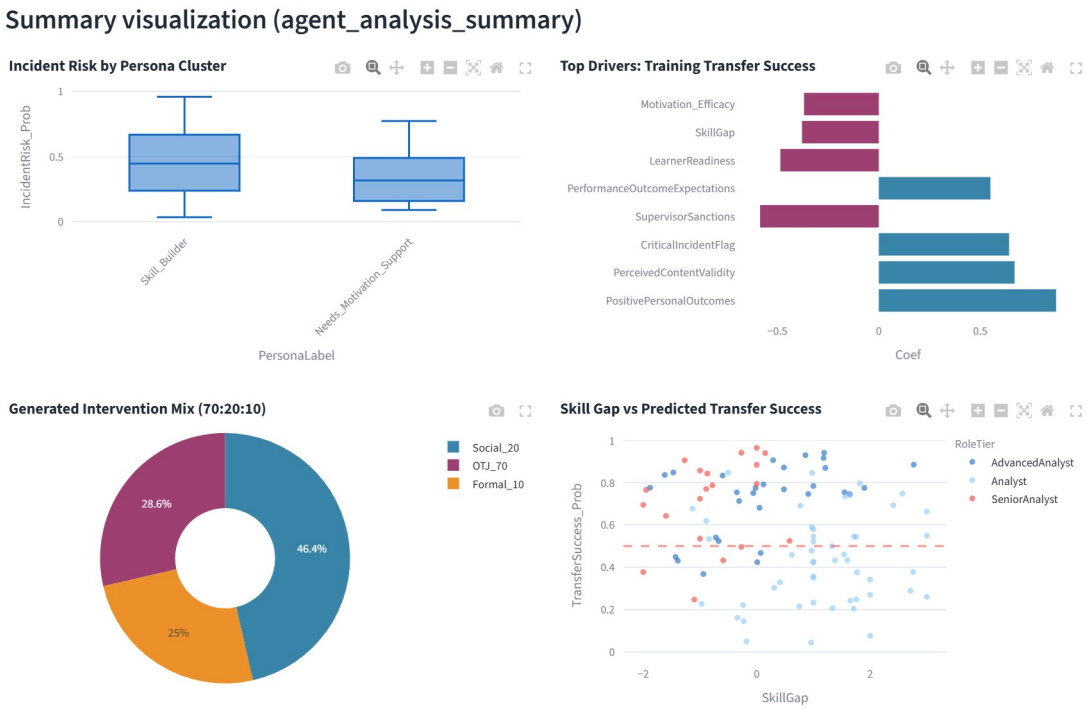

Figure 5: *Summary Visualisation module (agent_analysis_summary) from the live Streamlit application.* Top-left: Incident Risk by Persona Cluster—Skill_Builder analysts show higher median risk than Needs_Motivation_Support analysts. Top-right: Top Drivers for Training Transfer Success—Motivation_Efficacy and SkillGap dominate with expected directional signs, confirming H1 and H6. Bottom-left: Generated Intervention Mix (70:20:10)—Social_20 activities constitute 46.4% of total, reflecting the high proportion of risk-flagged cases requiring peer and managerial support. Bottom-right: Skill Gap vs. Predicted Transfer Success by role tier, showing the expected negative relationship.

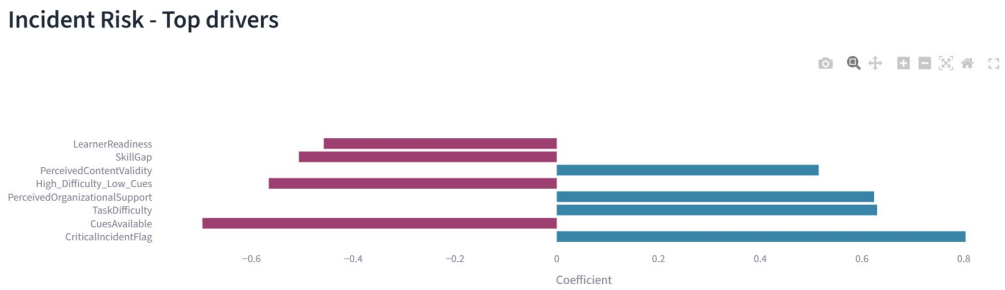

Figure 6: *Incident Risk — Top Drivers coefficient chart from the Model Performance and Theory Validation module.* CriticalIncidentFlag (+0.80), TaskDifficulty (+0.62), and PerceivedOrganisationalSupport (−0.61) are the dominant drivers. Purple bars indicate negative coefficients; teal bars indicate positive coefficients. All directional signs are consistent with hypotheses R1–R5 (Table 3), confirming ante-hoc theory alignment.

AnalystID	AoPID	Persona	IncidentRisk	SkillGap	OTJ_Activities	Social_Activities	Formal_Activities	Artifacts	KPI_Target
ANALYST_000	AOP_04	Skill_Builder	0.656	1.56	1	2	1	Observation_Checklist_V2, Critical_Incident_Analysis_Table, eLearning_Scenario_Modi	25% in 30 days
ANALYST_000	AOP_05	Skill_Builder	0.776	1.82	1	2	1	Observation_Checklist_V2, Critical_Incident_Analysis_Table, eLearning_Scenario_Modi	50% in 30 days
ANALYST_002	AOP_02	Skill_Builder	0.916	0.98	1	1	1	Observation_Checklist_V2, Critical_Incident_Analysis_Table, eLearning_Scenario_Modi	50% in 30 days
ANALYST_002	AOP_03	Skill_Builder	0.767	-1.13	0	1	0	Critical_Incident_Analysis_Table	50% in 30 days
ANALYST_002	AOP_05	Skill_Builder	0.601	2.41	2	1	1	Observation_Checklist_V2, PST_Cues_Strategies, Critical_Incident_Analysis_Table, eLe	25% in 30 days
ANALYST_003	AOP_01	Needs_Motivation_Support	0.773	-0.89	1	1	0	PST_Cues_Strategies, Critical_Incident_Analysis_Table	50% in 30 days
ANALYST_003	AOP_02	Needs_Motivation_Support	0.771	0.97	1	1	1	Observation_Checklist_V2, Critical_Incident_Analysis_Table, eLearning_Scenario_Modi	50% in 30 days
ANALYST_004	AOP_05	Skill_Builder	0.82	1.19	1	2	1	Observation_Checklist_V2, Critical_Incident_Analysis_Table, eLearning_Scenario_Modi	50% in 30 days
ANALYST_005	AOP_02	Skill_Builder	0.742	-1	0	1	1	Critical_Incident_Analysis_Table, eLearning_Scenario_Module	50% in 30 days
ANALYST_005	AOP_03	Skill_Builder	0.741	-1.27	0	1	0	Critical_Incident_Analysis_Table	50% in 30 days

Figure 7: *Intervention output table from the live system.* Each row is one analyst-AoP pair showing Persona, IncidentRisk, SkillGap, activity counts (OTJ/Social/Formal), evidence artifacts, and KPI target. ANALYST_002 / AoP_02 (IncidentRisk = 0.916; SkillGap = 0.98) receives the highest-priority plan with a 50% incident reduction target. All plans proceed to the Decision Queue before any deployment action.

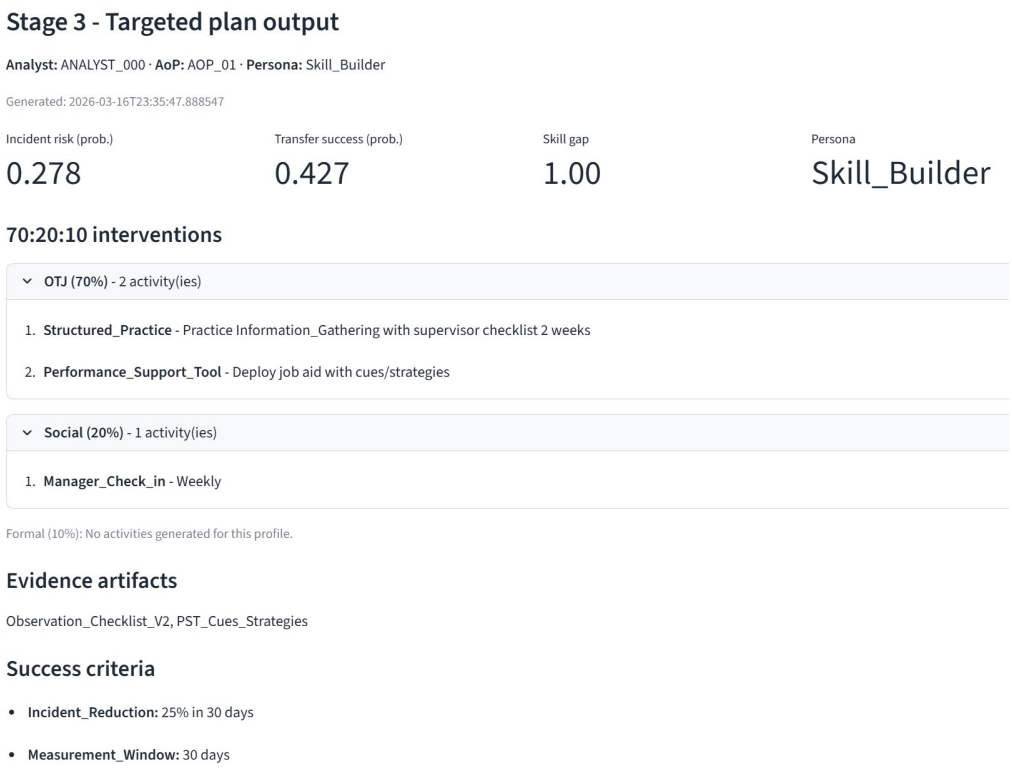

Figure 8: *Stage 3 — Targeted Plan Output for ANALYST_000 / AoP_01 (Skill_Builder; IncidentRisk* = 0.278*; TransferSuccess* = 0.427*; SkillGap* = 1.00*).* The system generated on 2026-03-16 and the plan remains PENDING. Recommended plan: OTJ (70%)—Structured Practice with supervisor checklist (2 weeks) and Performance_Support_Tool with cues/strategies; Social (20%)—Manager_Check-in weekly; Formal (10%)—no activities for this profile. Evidence artifacts: Observation_Checklist_V2, PST_Cues_Strategies. Success criterion: 25% incident reduction in 30 days. This screenshot confirms the human-in-the-loop governance described in Section 4.4.

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A System Architecture and Implementation

A.1 Technology Stack and Repository

Python 3.9+; scikit-learn $\geq$ 1.3; XGBoost $\geq$ 1.7; Plotly $\geq$ 5.15; Pydantic $\geq$ 2.0; Streamlit $\geq$ 1.28. Code: https://github.com/Zmugha1/dr-data-training-agent.

A.2 Repository Modules

`agents/` (9 specialised agents: data prep, risk prediction, clustering, intervention generation); `human_loop/` (Decision Queue, audit trail); `core/` (Pydantic schemas, state management); `pipeline/` (ML pipeline); `ui/` (Streamlit dashboard); `config/` (`auto_approve_interventions: false`, risk thresholds); `data/` (seed data, synthetic datasets, JSON audit files).

B Full Cross-Validation Results

Table 4: *Complete 5-fold stratified cross-validation AUC.*

Model	Target	F1	F2	F3	F4	F5	Mean	SD
LogReg	Transfer	0.841	0.837	0.844	0.835	0.839	0.839	**0.003**
	Incident	0.824	0.819	0.826	0.817	0.822	0.822	**0.004**
RF	Transfer	0.892	0.812	0.856	0.789	0.877	0.845	0.042
	Incident	0.871	0.798	0.839	0.776	0.865	0.830	0.041
XGBoost	Transfer	0.901	0.821	0.868	0.805	0.883	0.856	0.038
	Incident	0.885	0.809	0.851	0.792	0.878	0.843	0.037

Coefficient of variation (CV $= \sigma/\mu$): Theory-constrained: 0.004 (Transfer), 0.005 (Incident). Weak bias: $\sim$0.050. This 10–14$\times$ variance reduction occurs because tree-based models search for non-existent non-linearities in linearly-generated data.

C Synthetic Data Generation: Full Protocol

Role proportions: Analyst 50%, Advanced 35%, Senior 15%. Base proficiency: Analyst = 1.5, Advanced = 2.5, Senior = 3.5. LTSI scores: Truncated Normal$(\mu_{\text{tier}}, \sigma = 0.6, [1, 5])$. Both DGP intercepts $\beta_0 = \gamma_0 = 0$; class balance via `class_weight='balanced'`. Random seed: 42. Train/test split: 80/20 stratified. CV folds: 5-fold stratified. Total features: 23 (16 LTSI constructs + 7 engineered). Full DGP equations in Section 4.1.

D 70:20:10 Intervention Mapping Logic

On-the-Job (70%): SkillGap $> 0.5 \rightarrow$ Structured Practice + Observation Checklist; CuesAvailable $< 0.5 \rightarrow$ Performance Support Tool + Job Aid.

Social (20%): CriticalIncidentFlag = 1 or IncidentRisk $> 0.6 \rightarrow$ Peer Coaching + Incident Analysis Table; SupervisorSupport $< 3.0 \rightarrow$ Manager Check-in (weekly).

Formal (10%): TaskDifficulty $\geq 3 \rightarrow$ Scenario-Based eLearning (30 min).

KPI Targets: High Risk (> 0.7): 50% incident reduction in 30 days. Moderate Risk: 25% incident reduction in 30 days.

E Ethics and Production Deployment Limitations

All demonstration data is synthetically generated; no real employee data is used. The system includes UI disclaimers warning that synthetic linear data structurally favours linear models. Production limitations: real relationships may be non-linear; $k = 3$ requires validation; rules are deterministic; temporal validation is required. The human-in-the-loop architecture is designed to preserve manager accountability and prevent automated deployment under all conditions.